

Chapter 5: Force & Pressure

Class 8 Science • Simplified, Illustrated & Fun to Learn!

1. What is Force? 🚀

Definition: In science, a **push** or a **pull** applied to an object is called a force. A force always requires an *interaction* between two objects!

SI Unit: Newton (N) **IMPORTANT**

Effects of Force: Applying a force can cause:

- ✦ A stationary object to start moving.
- ✦ A moving object to change its speed (go faster or slower).
- ✦ A moving object to change its direction of motion.
- ✦ A change in the shape or size of an object.

2. Types of Forces 🤝⚡

A. Contact Forces 🖐️

Requires physical contact between objects.

1. Muscular Force:

- ✦ Created by muscles expanding & contracting.
- ✦ *Examples:* Chewing food, pumping blood, lifting a heavy school bag.

2. Frictional Force:

- ✦ Acts opposite to the moving direction.
- ✦ Slows down or stops motion.
- ✦ Caused by tiny surface roughness.

💡 **Streamlined Design:** Fluids (air & water) also exert friction! Planes, boats, and high-speed trains have special smooth shapes to cut through drag easily.

B. Non-Contact Forces ⚡

Acts from a distance without touching!

1. Magnetic Force:

- ✦ Exerted by magnets on metals or other magnets.
- ✦ Like poles repel (N - N); Opposite poles attract (N - S).

2. Electrostatic Force:

- ✦ Force exerted by static electrical charges.
- ✦ Rubbing items generates static charges!

3. Gravitational Force:

- ✦ The Earth's natural pull on every object towards its center.

3. Mass vs. Weight

Property	Weight (W) 	Mass (m) 
Meaning	Force by which gravity pulls an object.	Total amount of matter inside an object.
Value	Changes with location & gravity.	Stays the same everywhere in the universe!
Unit	Newton (N)	Kilogram (kg) / Gram (g)
Formula	$W = \text{mass} \times \text{gravity}$	N/A (Fixed quantity)

 **Science Alert:** Using weight and mass interchangeably in daily talk is common, but scientifically they are completely different!

4. Spring Balance & Smallest Values

A **Spring Balance** measures an object's weight. Hanging a weight stretches the internal spring, indicating the force on a scale marked in Newtons (**N**) and grams (**g**).

How to find the Least Count (Smallest Value):

LEAST COUNT FORMULA

$$\text{Least Count} = \text{Value Difference} / \text{Number of Divisions}$$

Quick Calculation Example:

$$\text{If 5 divisions exist between 0 N and 1 N} \rightarrow \text{Least Count} = 1 \text{ N} / 5 = 0.2 \text{ N}$$

5. Floating, Upthrust & Archimedes

Upthrust (Buoyant Force): An upward push applied by liquids on any object placed in them, working directly against gravity!

 **Will it Float or Sink?**

- ✦ If Upthrust > Gravity (Weight) → Object **FLOATS** 
- ✦ If Upthrust < Gravity (Weight) → Object **SINKS** 

Archimedes' Principle:

"When an object is dipped in water, it feels an upward force equal to the exact weight of the water displaced by it!"



Fun Fact (Pumice Stone): Pumice is a volcanic rock full of tiny air pockets! Because of these trapped air bubbles, it displaces enough water to produce a strong upward buoyant force, allowing this stone to float on water!